Week 8: 19 – 23 Oct 2026

Subject: Mathematics **Unit Title:** Geometry/Statistics **Grade :** 12DP

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1 Frequency Tables	• I can analyze data by organizing it into frequency tables, calculating measures of center, and interpreting relative and cumulative frequency graphs to identify patterns and trends.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.1 Make goals clear, visible, and supported by short-term objectives.	Reflection prompts, notebooks The resources will be available on student's Teams channel Revision Village problems based on variations (direct and indirect) and geometry of 3D shapes. https://www.revisionvillage.com/ Past IB AI SL questions on Statistics. https://www.savemyexams.com/members/ InThinking online assessment to measure student progress on the concept understanding of Statistics. https://thinkib.net/mathapplications
Lesson 2 Direct Variation	• I can solve and model directly proportional relationships by identifying the constant of variation and applying direct variation equations to real-world situations.		
Lesson 3 Indirect Variation	• I can solve and model inversely proportional relationships by determining the constant of variation and applying inverse variation equations to real-world situations.		
Lesson 4 Formative	• I can analyze relationships in data by organizing information in frequency tables, interpreting graphs, and modeling direct and inverse variation to solve real-world problems.		
Teacher's notes/ Reminders ملاحظات المعلم والتذكير			

Technology Integration	Use Desmos, GeoGebra and Excel for data visualization, boxplots and frequency tables; Teams for sharing and reflection.

دمج التكنولوجيا	
Cross-curricular links الروابط بين المواد الدراسية	Science (data analysis of experiments), Economics (market sampling), Geography (population data comparison).
External Assessment Integration دمج التقييمات الخارجية	All content is covered in IB Past Exams
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Wellbeing:** • Confidence in solving real-world maths problems. • Resilience when learning from mistakes. • Collaboration through shared investigations and discussions. • Self-awareness by setting and reflecting on learning goals. **IB ATL Skills:** • **Thinking Skills** – Interpret numerical data to identify patterns and trends. • **Communication Skills** – Present and explain statistical findings clearly. • **Self-Management Skills** – Plan and execute data collection and organization. • **Research Skills** – Use authentic data sources to analyze results. • **Collaboration Skills** – Work in groups to design surveys and interpret results.
Standards المعايير	IB Math AI SL Standards: • 7.SP.A.1-2 • 7.SP.B.3 • HSS-ID.A.1-2 • 8.EE.B.5-6 • 8.F.B.4 • HSA-CED.A.2 • HSF-BF.A.1
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can we trust data to tell the truth if the method of collecting it is biased?
Success Criteria معايير النجاح (Measurable)	Lesson 1: ▨ **All Students Must:** Create frequency tables and calculate mean and median from frequency data. ◍ **Most Students Should:** Interpret relative and cumulative frequency graphs and describe patterns in the data. ▲ **Some Students Could:** Use frequency data and graphs to make predictions, compare data sets, and justify conclusions.

	Lesson 2: ▨ **All Students Must:** Identify direct variation equations and determine the constant of variation. ◍ **Most Students Should:** Solve problems involving direct variation and explain the relationship between variables. ▲ **Some Students Could:** Create and defend a mathematical model using direct variation to represent and solve a real-world problem. Lesson 3: ▨ **All Students Must:** Identify inverse variation equations and determine the constant of variation. ◍ **Most Students Should:** Solve problems involving inverse variation and explain how changes in one variable affect the other. ▲ **Some Students Could:** Solve problems involving inverse variation and explain how changes in one variable affect the other. Lesson 4: ▨ **All Students Must:** Create frequency tables and calculate measures of center from frequency data. Identify direct and inverse variation equations. Determine constants of variation and solve routine problems. ◍ **Most Students Should:** Interpret relative and cumulative frequency graphs. Compare direct and inverse variation relationships. Use equations, tables, and graphs to explain patterns and solve contextual problems. ▲ **Some Students Could:** Develop mathematical models from real-world situations involving variation and data analysis. Evaluate the reasonableness of solutions and justify conclusions using multiple representations and statistical evidence.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Belonging الإنتماء Respect الاحترام Heritage التراث